C2 Amount of Substance – Knowledge Organiser

AQA A-level Chemistry • Set 1: Moles and Formulae **Papers 1 & 2**

#	Question	Answer
1	Avogadro constant (N_A)	6.022×10^{23} mol^{-1}
2	Mole	Amount containing 6.022×10^{23} particles
3	Molar mass	Mass of 1 mol of a substance (g mol^{-1})
4	Equation linking moles, mass and M_r	$n = m \div M_r$
5	Equation linking moles, concentration and volume	$n = c \times V$ (V in dm^3)
6	Equation for number of particles	$N = n \times N_A$
7	Convert cm^3 to dm^3	÷ 1000
8	Units of concentration	mol dm^{-3}
9	Empirical formula	Simplest whole-number ratio of atoms of each element
10	Molecular formula	Actual number of atoms of each element in a molecule

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AQA A-level Chemistry • Set 2: Ideal Gas Equation **Papers 1 & 2**

#	Question	Answer
1	Ideal gas equation	$pV = nRT$
2	Units of p	Pa
3	Units of V	m^3
4	Units of T	K
5	Value of R	8.31 J K^{-1} mol^{-1}
6	Convert °C to K	+ 273
7	Convert cm^3 to m^3	× 10^{-6}
8	Convert dm^3 to m^3	× 10^{-3}
9	Convert kPa to Pa	× 10^3
10	Molar gas volume at room temperature and pressure	24 dm^3 mol^{-1}

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AQA A-level Chemistry • Set 3: Yield, Atom Economy and Titrations **Papers 1 & 2**

#	Question	Answer
1	Percentage yield equation	Actual yield ÷ theoretical yield × 100
2	Atom economy equation	M_r of desired product ÷ total M_r of reactants × 100
3	Benefit of high atom economy	Less waste
4	Limiting reactant	Reactant completely used up
5	Reactant in excess	Reactant left over at the end
6	Standard solution	Solution of known concentration
7	Concordant titres	Within 0.10 cm^3 of each other
8	Precision of burette readings	Nearest 0.05 cm^3
9	Ionic equation for neutralisation	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$
10	Spectator ions	Ions that do not take part in the reaction

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AQA A-level Chemistry • Set 4: Water of Crystallisation and Uncertainty **Papers 1 & 2**

#	Question	Answer
1	Water of crystallisation	Water molecules bonded within a crystal structure
2	Hydrated	Contains water of crystallisation
3	Anhydrous	Contains no water of crystallisation
4	Percentage by mass of an element	$(A_r \times \text{number of atoms}) \div M_r \times 100$
5	Convert mol dm^{-3} to g dm^{-3}	$\times M_r$
6	Percentage uncertainty equation	Uncertainty ÷ measured value × 100
7	Uncertainty of one burette reading	±0.05 cm^3
8	Uncertainty of a titre	±0.10 cm^3
9	Number of readings in a titre	2
10	Glassware used to make up a standard solution	Volumetric flask